BISHOW LAMSAL

Agentic AI | Generative AI | Machine Learning Engineer

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LinkedIn | GitHub | Kaggle | Portfolio

CAREER OBJECTIVE

Production-focused AI/ML Engineer with 4+ years building and deploying AI systems across agentic automation, computer vision, NLP, and generative AI. Delivered end-to-end solutions for global clients including the UN FAO and US financial sector — from agentic QA platforms with Human-in-the-Loop controls and multi-model fallback to real-time edge vision systems. Seeking to build intelligent, high-impact systems across fintech, healthcare, and autonomous domains.

TECHNICAL SKILLS

Languages: Python, TypeScript
AI / ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Hugging Face, Transformers, Pandas, NumPy
LLMs & Agents: OpenAI, Claude, LLaMA, Phi, Qwen, RAG, MCP, LangChain, LangGraph, Playwright
Databases & Vector Stores: MongoDB, Redis, Qdrant, MEM0, SQL
Deployment & Serving: FastAPI, Flask, gRPC, Streamlit, Docker, Jenkins, AWS, Nepali Cloud, CI/CD
Monitoring & Observability: Grafana, Prometheus, Loki, cAdvisor, TensorBoard, WandB
Edge AI & GPU: CUDA, TensorRT, cuDNN, Jetson Xavier

WORK EXPERIENCE

Machine Learning Engineer — **Vertex Special Technology** Jan 2024 – Present

Built and shipped multiple production AI systems for international clients — spanning agentic QA automation, financial document intelligence, and UN procurement automation — as the core AI/ML software engineer driving end-to-end delivery.

- Architected and led development of AI Stellar, an agentic QA automation platform using Playwright and LLM-driven UI agents that generates hundreds of reusable test scenarios from a single user prompt, achieving 5x to 10x faster test execution compared to manual testing workflows.
- Built a gRPC-based chat interface that interprets user intent and dynamically routes between UI automation and test case generation workflows with context-aware decision making.
- Integrated MCP (Model Context Protocol) to connect automation tools seamlessly across agent workflows, enabling flexible and extensible tool use within the agentic pipeline.
- Designed a DOM-caching strategy that eliminates redundant AI calls for repeat or modified test cases, significantly cutting automation cost and latency compared to conventional AI-heavy QA tools.
- Designed an efficient memory architecture with dedicated short-term and long-term memory layers between agent nodes, ensuring each agent is supplied with the right context at the right time and enabling coherent multi-step reasoning across complex automation workflows.
- Implemented Human-in-the-Loop (HITL) intervention points that detect low-confidence or ambiguous inputs, pause agent execution, prompt the user for clarification, and rerun the pipeline with validated context.
- Engineered a full CI/CD pipeline specifically for the AI Stellar agentic platform with Grafana, Prometheus, Loki, and cAdvisor for real-time monitoring across all deployed agents.
- Integrated multi-model fallback with Claude for reliability along with prompt versioning, model versioning, and detailed token usage and pricing tracking per run.
- Delivered an end-to-end agentic procurement system for the UN FAO automating the full RFQ and ITB workflow including document reading, data extraction, structured analysis, spreadsheet generation, and LPC report creation, deployed within

Microsoft Teams using Microsoft Power Automate and LangGraph, enabling a single operator to replace what previously required multiple staff.

- Built ScAInit, a financial document intelligence system combining layout models, OCR engines, and fine-tuned VLMs including Phi and Qwen, achieving 95% layout accuracy on financial bank data and 90% table structure recognition across complex borderless tables and multi-span rows and columns.

Machine Learning Engineer Associate — Bottle Technology Sep 2022 – Sep 2023

Designed and deployed real-time computer vision systems on live city infrastructure across Kathmandu and Lalitpur, delivering traffic analytics and crowd management solutions on edge hardware.

- Developed and deployed a real-time traffic analytics system across 4 cameras at 4 locations in Kathmandu and Lalitpur, processing live footage at 20 FPS on Jetson Xavier edge devices.
- Implemented vehicle detection and tracking using YOLO and DeepSORT with geometric and motion-based logic to detect one-way traffic violations and illegal highway stops.
- Optimized model inference via TensorRT, significantly improving GPU utilization and achieving real-time 20 FPS throughput on constrained edge hardware.
- Analyzed over 100 hours of recorded footage to deliver actionable traffic insights for Kathmandu and Lalitpur Metropolitan city authorities.
- Conducted real-time crowd management analysis at BNB Hospital and ticket counters, tracking and counting large volumes of individuals to support operational planning.
- Trained specialized OCR models for Devanagari script license plate recognition and Nepali ID card information extraction, enabling accurate reading of Nepali-language documents.
- Built and deployed end-to-end ML pipelines from model design to cloud integration using Docker, Jenkins, and AWS.

Machine Learning Trainee — Bottle Technology Jun 2022 – Aug 2022

Built foundational skills in applied computer vision — implementing algorithms from research papers, annotating data, and benchmarking detection models across speed and accuracy trade-offs.

- Studied and implemented classical image processing algorithms from scratch, both with and without OpenCV, based on research papers.
- Annotated training data, fine-tuned models, and tracked performance across experiments using TensorBoard and Weights and Biases.
- Researched and benchmarked multiple versions of YOLO for real-time object detection, comparing performance across speed and accuracy metrics.
- Explored supervised and unsupervised techniques for image color analysis including RGB color spaces, grayscale conversion weights, and image binarization.

PERSONAL PROJECTS

Healthcare Data Extraction — Cloud-based PDF health records pipeline

- Built a pipeline to process approximately 1TB of split PDF health records, identifying redundant files, detecting important regions, and extracting key-value pairs and table data into structured JSON stored in MongoDB using a rule-based approach with zero patient data sent to LLMs, improving privacy and significantly reducing operational costs.
- Developed a Streamlit UI for patient data retrieval that processes thousands of files in minutes without any LLM dependency.

Face Detection, Recognition & Data Extraction — Computer vision identity pipeline

- Built a face similarity detection system combined with document classification and structured data extraction using a multi-engine OCR pipeline including PaddleOCR, TrOCR, and EasyOCR for maximum accuracy across varied document formats.
- Deployed on Nepali Cloud to ensure complete data sovereignty, keeping all sensitive identity data within the country's infrastructure.

Advanced E-Video-KYC — Real-time liveness authentication system

- Developed a real-time liveness detection system recognizing facial movements including blink, mouth open, and head turns, integrated with OCR-based document data extraction and voice assistant support.

Personal Expense Tracker — Cross-platform mobile finance app (Flutter / Kotlin / Python)

- Building a mobile application that automatically extracts transaction data from bank SMS notifications using NLP-based parsing, categorizes income and expense flows, and visualizes spending patterns through interactive dashboards with multi-bank account management and budget estimation.

News Recommendation System — Hybrid News Recommendation Engine with Multi-View Attention

- Built a hybrid recommendation engine using the MIND dataset combining content-based, collaborative filtering, and popularity-based approaches with Word2Vec embeddings and Multi-View Attention.

Real-time Work Surveillance — AI-Powered Employee Activity Monitoring

- Developed employee activity monitoring system with real-time face detection and recognition using vector databases, tracking seated hour, mobile usage and other behaviours with web-based visualization.

ACADEMIC QUALIFICATION

Bachelor of Engineering in Computer Science — Visvesvaraya Technological University, Bangalore, India

CGPA: 7.53 / 10 | Aug 2016 – Dec 2020

CERTIFICATIONS & TRAINING

- Mentored participants at Hack4SafeFood Hackathon
- 4-Month Data Science Training, Deerwalk Training Center
- 7-Week Web Development Internship, Rbits Technology
- Active Kaggle competitor working with datasets from Kaggle and UCI Repository